

Online Appendix for

Causal Inference with Ordinal Outcomes: Flexible Estimation Framework Using Normalizing Flows

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This appendix provides supplementary theoretical derivations, estimation details, extensions to heterogeneous reporting thresholds, comprehensive simulation designs with full results, and detailed tables for the empirical applications.

Contents

A	Additional Theoretical Details	3
A.1	Probability-Based Estimands and Latent Scale Invariance	3
A.2	Wasserstein Distance as a Distributional Metric	3
A.3	Nonparametric Identification of the Latent Error Distribution	3
B	Extensions: Heterogeneous Reporting Thresholds	6
C	Estimation Details and Proofs	7
C.1	Architectural Details of the Flow Estimators	7
C.1.1	The Structured Flow Model	7
C.1.2	The Model-Free Flow Model	8
C.2	Warm-Start Initialization and Two-Stage Optimization	8
C.2.1	Probit Warm-Start Initialization	8
C.2.2	Two-Stage Training and Split Learning Rates	8
C.2.3	Second-Stage L-BFGS Polishing	9
C.3	Theoretical Consistency and Bootstrap Validity	10
D	Comprehensive Simulation Details & Full Results	12
D.1	Detailed Specifications of the Seven Data-Generating Processes	12
D.2	Full Simulation Results Table	12

E	Empirical Application Details & Full Tables	14
E.1	Mattingly et al. (2025): Persuasive Effects of Authoritarian Propaganda	14
E.2	Tomz and Weeks (2020): Human Rights and Public Support for Intervention	14

A. Additional Theoretical Details

A.1. Probability-Based Estimands and Latent Scale Invariance

Recall that the observed ordinal response $Y_i \in \{0, 1, \dots, J\}$ is generated via the threshold-crossing model:

$$Y_i = j \iff \alpha_{j-1} < Y_i^* \leq \alpha_j, \quad (\text{A.1})$$

where Y_i^* is the latent potential outcome and $-\infty = \alpha_{-1} < \alpha_0 < \dots < \alpha_J = \infty$.

Let $\mathcal{T} : \mathbb{R} \rightarrow \mathbb{R}$ be a strictly increasing, positive affine transformation: $\mathcal{T}(y) = a + cy$ with $c > 0$. Under this transformation, we define the transformed latent outcome $\tilde{Y}_i^* = \mathcal{T}(Y_i^*)$ and the transformed thresholds $\tilde{\alpha}_j = \mathcal{T}(\alpha_j)$. The observed category probabilities under the transformed scale are given by:

$$\begin{aligned} P(\tilde{Y}_i = j) &= P(\tilde{\alpha}_{j-1} < \tilde{Y}_i^* \leq \tilde{\alpha}_j) \\ &= P(a + c\alpha_{j-1} < a + cY_i^* \leq a + c\alpha_j) \\ &= P(\alpha_{j-1} < Y_i^* \leq \alpha_j) = P(Y_i = j). \end{aligned}$$

Because the category probabilities $\pi_j(d) = P(Y_i(d) = j)$ remain unchanged under any positive affine scale shift, any causal quantity defined directly as a functional of these probabilities is strictly invariant to the choice of scale.

A.2. Wasserstein Distance as a Distributional Metric

To summarize overall treatment effects across the entire distribution without relying on arbitrary numerical score assignments, we employ the one-dimensional L_1 Wasserstein distance (also known as the Earth Mover's Distance). For two cumulative distribution functions (CDFs) F_1 and F_0 defined over the ordered categories $\{0, 1, \dots, J\}$, the Wasserstein distance is formulated as:

$$\mathcal{W}(F_1, F_0) = \sum_{j=0}^{J-1} |F_1(j) - F_0(j)|, \quad (\text{A.2})$$

where $F_d(j) = P(Y_i(d) \leq j)$ for $d \in \{0, 1\}$. This metric measures the minimum cost of shifting probability mass to transform the control outcome distribution into the treatment outcome distribution, preserving the natural ordering of categories.

A.3. Nonparametric Identification of the Latent Error Distribution

A primary theoretical contribution of the conditional normalizing flow framework is the relaxation of parametric assumptions on the latent error distribution. To establish that this relaxation is theoretically sound, we formally prove that the latent error distribution F and the thresholds α are nonparametrically identified from the observed data.

Let the latent potential outcome be represented by the single-index structure:

$$Y_i^* = m(X_i, D_i) + \varepsilon_i, \quad (\text{A.3})$$

where $m(X_i, D_i) \in \mathbb{R}$ is the systematic index, and ε_i is a continuous latent disturbance with cumulative

distribution function F . Let $V_i = m(X_i, D_i)$ denote the value of the index for individual i .

We impose the following standard regularity assumptions:

ASSUMPTION A.1 (Exogeneity). *The latent error is statistically independent of the treatment and covariates: $\varepsilon_i \perp (X_i, D_i)$.*

ASSUMPTION A.2 (Index Identification). *The systematic index $V_i = m(X_i, D_i)$ is identified up to a positive affine transformation from the conditional mean of the observed outcomes.*

ASSUMPTION A.3 (Continuous Large Support). *The systematic index V_i is a continuous random variable with support $\mathcal{V} \subseteq \mathbb{R}$. For each $j \in \{0, \dots, J-1\}$, the set of shifted values $\{\alpha_j - v : v \in \mathcal{V}\}$ contains the support of the latent error ε_i .*

ASSUMPTION A.4 (Normalizations). *To fix the location and scale of the latent variable, we normalize the latent space by setting the first threshold $\alpha_0 = 0$ (location normalization) and regulating the model parameters of the systematic index $m(X_i, D_i)$ to have an L_2 norm of 1.*

Under these standard normalizations, the location and scale of the latent variable are uniquely pinned down. Consequently, the systematic index $V_i = m(X_i, D_i)$ is fully point-identified from the conditional mean of the observed outcomes (Manski, 1988; Klein and Spady, 1993). We can now establish the nonparametric identification of the latent error distribution and the remaining thresholds:

THEOREM A.1 (Nonparametric Identification). *Under Assumptions A.1–A.4, the latent error distribution F and the partition thresholds α_j for $j \in \{1, \dots, J-1\}$ are uniquely and nonparametrically identified.*

PROOF. By the threshold-crossing model, the observed cumulative probability of responding at or below category j conditional on the index value $V_i = v$ is:

$$\begin{aligned} P(Y_i \leq j \mid V_i = v) &= P(Y_i^* \leq \alpha_j \mid V_i = v) \\ &= P(V_i + \varepsilon_i \leq \alpha_j \mid V_i = v) \\ &= P(\varepsilon_i \leq \alpha_j - v \mid V_i = v). \end{aligned} \tag{A.4}$$

Applying Assumption A.1 (independence), this conditional probability simplifies to:

$$P(Y_i \leq j \mid V_i = v) = F(\alpha_j - v). \tag{A.5}$$

First, we identify the latent error distribution $F(\cdot)$. Under the location normalization in Assumption A.4, we have $\alpha_0 = 0$. Setting $j = 0$ in Equation (A.5) yields:

$$P(Y_i \leq 0 \mid V_i = v) = F(-v). \tag{A.6}$$

By Assumption A.3 (large support), the index V_i varies continuously over $\mathcal{V}$. Therefore, we can recover the value of $F(u)$ for any u in the support of ε_i by evaluating the observed probability at $v = -u$:

$$F(u) = P(Y_i \leq 0 \mid V_i = -u). \tag{A.7}$$

Because the left-hand side is an observed conditional probability, the function $F(\cdot)$ is uniquely identified on its entire support.

Second, we identify the remaining thresholds α_j for $j \geq 1$. For any $j > 0$, we have:

$$P(Y_i \leq j \mid V_i = v) = F(\alpha_j - v). \quad (\text{A.8})$$

Since $F(\cdot)$ is a continuous and strictly monotonic cumulative distribution function, its inverse $F^{-1}(\cdot)$ exists and is uniquely identified. Applying the inverse function to both sides yields:

$$\alpha_j - v = F^{-1}(P(Y_i \leq j \mid V_i = v)), \quad (\text{A.9})$$

which identifies the threshold α_j as:

$$\alpha_j = v + F^{-1}(P(Y_i \leq j \mid V_i = v)). \quad (\text{A.10})$$

Because this relation holds for any $v \in \mathcal{V}$, the thresholds α_j are overidentified and uniquely recovered, completing the proof. $\square$

The continuous large support assumption (Assumption A.3) is the key theoretical requirement that enables nonparametric identification. Intuitively, it ensures that the systematic index V_i has enough variation to "sweep" the fixed thresholds across the entire range of the latent attitude, allowing us to map out the exact shape of the latent error distribution F without relying on any parametric assumptions.

B. Extensions: Heterogeneous Reporting Thresholds

The threshold-crossing model in Section 2.1 assumes that all respondents share common thresholds α_j . In many survey settings, respondents may interpret the ordinal response scale differently based on their individual characteristics, a phenomenon known as differential item functioning (DIF).

We can extend the proposed flexible estimation framework to accommodate heterogeneous reporting behavior. Following the generalized ordered response framework (Williams, 2006), we model the individual-specific thresholds $\alpha_{j,i}$ as a function of respondent-level covariates $W_i \subseteq X_i$:

$$\alpha_{j,i} = \alpha_j + W_i^\top \gamma_j, \quad (\text{B.1})$$

where γ_j is a vector of parameters capturing how the thresholds shift across individuals, subject to the ordering constraint:

$$\alpha_{0,i} < \alpha_{1,i} < \cdots < \alpha_{J-1,i} \quad \forall i. \quad (\text{B.2})$$

Under this specification, the probability of observing category j becomes:

$$\begin{aligned} P(Y_i = j \mid D_i, X_i) &= F_\phi \left(\alpha_j + W_i^\top \gamma_j - h(X_i) - D_i^\top \tau \mid D_i, X_i \right) \\ &\quad - F_\phi \left(\alpha_{j-1} + W_i^\top \gamma_{j-1} - h(X_i) - D_i^\top \tau \mid D_i, X_i \right). \end{aligned} \quad (\text{B.3})$$

This generalized formulation is fully compatible with both the semiparametric KDE-based estimator and the conditional normalizing flow framework. In the flow-based setting, the parameters γ_j are jointly estimated via maximum likelihood alongside the flow parameters ϕ and the index coefficients.

C. Estimation Details and Proofs

This section provides the complete architectural, optimization, and algorithmic details of the proposed conditional normalizing flow estimators implemented in the paper, followed by formal proofs of asymptotic consistency and bootstrap validity.

C.1. Architectural Details of the Flow Estimators

The framework implements two primary models: the *Structured Flow* and the *Model-Free Flow*.

C.1.1. The Structured Flow Model

The Structured Flow model retains the interpretability of a linear single-index latent variable model while learning the latent error distribution nonparametrically. To ensure formal identification, we impose strict mathematical constraints on the parameter space:

- a. **Unit L2-Norm Index Scale Normalization:** To resolve the scale non-identification of the latent index coefficients (Theorem 1), as in the Assumption A.4 we normalize the model parameter vectors of the systematic index $m(X_i, D_i)$, $\beta \in \mathbb{R}^p$ to lie on the unit sphere during training. Given a raw trainable parameter vector $\theta \in \mathbb{R}^p$, we analytically enforce:

$$\beta = \frac{\theta}{\max(\|\theta\|_2, 10^{-12})}. \quad (\text{C.1})$$

- b. **Recursive Threshold Ordering Constraint:** To prevent threshold crossing (which would yield negative category probabilities), we parameterize the partition thresholds $\alpha \in \mathbb{R}^{J-1}$ recursively. The first threshold α_1 is anchored at the origin ($\alpha_1 = 0$) to fix location. For all subsequent thresholds $j \in \{2, \dots, J-1\}$, we define:

$$\alpha_{j,i} = \alpha_{j-1,i} + \text{softplus}(\lambda_{j-1,i}) + \delta_{\min}, \quad (\text{C.2})$$

where $\lambda_{j-1,i}$ is a raw parameter (which may be modeled as a linear function of threshold-specific covariates Z_i to allow for heterogeneous reporting behavior), and $\delta_{\min} > 0$ is a user-specified minimum threshold gap (set to 10^{-4} by default). This formulation mathematically guarantees that $-\infty < \alpha_{1,i} < \alpha_{2,i} < \dots < \alpha_{J-1,i} < \infty$ for all individuals.

- c. **Location-Scale Flow Alignment:** Let $E_{i,j} = \alpha_j - X_i\beta$ represent the boundary values in the latent error space. To facilitate stable optimization of the spline parameters, we apply a learnable location shift $b \in \mathbb{R}$ and scale parameter $a > 0$ to align the error boundaries prior to passing them through the flow:

$$\tilde{E}_{i,j} = \frac{E_{i,j} - b}{\text{softplus}(a_{\text{raw}})}, \quad (\text{C.3})$$

where a_{raw} is the raw scale parameter.

- d. **One-Dimensional Rational-Quadratic Spline (RQS) Mapping:** The aligned boundaries $\tilde{E}_{i,j}$ are transformed into standard normal latent coordinates $Z_{i,j}$ using a custom 1D monotonic rational-quadratic spline T_ϕ^{-1} :

$$Z_{i,j} = T_\phi^{-1}(\tilde{E}_{i,j}). \quad (\text{C.4})$$

The spline divides the space $[-B, B]$ (where the tail bound B is set to 10.0 by default) into $K = 16$ bins. Within each bin, T_ϕ^{-1} is parameterized as a rational-quadratic spline with analytical derivatives. Outside

the bounds, the transformation is strictly linear to ensure stable behavior at the tails.

Once the boundary mappings $Z_{i,j}$ are evaluated, the conditional probability for category $j \in \{0, \dots, J-1\}$ is calculated exactly using the standard normal cumulative distribution function $\Phi(\cdot)$:

$$P(Y_i = j \mid X_i, Z_i) = \Phi(Z_{i,j+1}) - \Phi(Z_{i,j}), \quad (\text{C.5})$$

where $Z_{i,0} = -\infty$ and $Z_{i,J} = \infty$.

C.1.2. The Model-Free Flow Model

The Model-Free Flow relaxes both the linear index and the parametric error assumptions. Instead of a scalar latent variable model, it directly parameterizes the threshold boundaries $B_{i,j} = \alpha_j$ and uses an autoregressive conditional flow to transform these thresholds into latent coordinate space $Z_{i,j}$:

$$Z_{i,j} = T_{\text{AR}}^{-1}(B_{i,j} \mid X_i), \quad (\text{C.6})$$

where T_{AR}^{-1} is implemented as a Masked Piecewise Rational Quadratic Autoregressive Transform conditioned on the full covariate context vector X_i . This formulation allows the conditional ordinal response distribution to adapt nonparametrically to complex, high-dimensional interaction and non-linear patterns.

C.2. Warm-Start Initialization and Two-Stage Optimization

To ensure high numerical stability and rapid training, we implement a highly structured training pipeline:

C.2.1. Probit Warm-Start Initialization

Before maximizing the exact flow likelihood, we initialize the parameters of the *Structured Flow* using standard Ordered Probit estimates obtained via BFGS from the `statsmodels` library. Let $\beta_{\text{probit}} \in \mathbb{R}^p$ and $\alpha_{\text{probit}} \in \mathbb{R}^{J-1}$ denote the fitted probit parameters. Because the flow model constrains $\|\beta\|_2 = 1$, we scale and map these estimates to initialize the flow parameters as follows:

- a. Compute the scale factor: $s = \|\beta_{\text{probit}}\|_2$.
- b. Initialize the index parameter: $\theta_{\text{init}} = \beta_{\text{probit}}$, which guarantees that the normalized vector β equals the normalized direction of the Probit coefficients.
- c. Initialize the scale parameters: $a_{\text{raw, init}} = \text{softplus}^{-1}(1/s)$, matching the variance scaling of the Probit.
- d. Initialize the location shift: $b_{\text{init}} = -\alpha_{\text{probit},0}/s$.
- e. If $J > 2$, initialize the gap parameters using the scaled differences:

$$\lambda_{j-1,\text{init}} = \text{softplus}^{-1} \left(\frac{\alpha_{\text{probit},j} - \alpha_{\text{probit},j-1}}{s} - \delta_{\min} \right). \quad (\text{C.7})$$

This analytical initialization ensures that the model starts training *exactly* at the Ordered Probit solution, guaranteeing that the flow can only improve upon the classical baseline.

C.2.2. Two-Stage Training and Split Learning Rates

Training proceeds in two distinct stages to protect the fragile threshold-gaps during early training:

- **Stage 1 (Epochs 1–20):** The spline parameters ϕ are completely frozen. The Adam optimizer ($lr = 5 \times 10^{-3}$) optimizes only the index θ , thresholds α , location b , and scale a . This allows the model to find a stable linear projection before warping the distribution.
- **Stage 2 (Epoch 21 onward):** The spline parameters ϕ are unfrozen. To prevent training instability, we use split learning rates: the index and threshold parameters continue optimizing at lr , while the spline parameters are optimized at a 10x smaller learning rate ($0.1 \times lr$).

C.2.3. Second-Stage L-BFGS Polishing

Once Adam meets the early stopping criteria (a patience of 15 epochs with a improvement tolerance of 10^{-4}), we transition to a second-stage L-BFGS optimizer. Using a Strong-Wolfe line search, L-BFGS leverages exact Hessian approximations to polish the parameters, driving the gradient of the exact log-likelihood precisely to zero and ensuring highly accurate convergence.

C.3. Theoretical Consistency and Bootstrap Validity

Let $\Theta = \{\beta, \alpha, a_{\text{raw}}, b, \phi\} \in \Theta$ collect all parameters of the Structured Flow model, and let $\ell_n(\Theta)$ denote the sample log-likelihood. Let F_0 denote the true latent error distribution, and let Θ_0 denote the true parameter vector. To establish consistency and asymptotic normality of the flow-based maximum likelihood estimator, we impose the following standard regularity conditions:

ASSUMPTION C.5 (Flow Approximation and Identification). *The flow family $\{F_\phi : \phi \in \Phi\}$, consisting of the distributions representable by the specified flow model, is sufficiently rich that there exists a parameter vector ϕ_0 in the closure of the parameter space such that $F_{\phi_0} = F_0$. More generally, F_{ϕ_0} may approximate the true latent error distribution F_0 arbitrarily well in Kullback–Leibler divergence.*

ASSUMPTION C.6 (Regularity Conditions). *The joint parameter space Θ is compact (or can be restricted to a compact subset). The density of the covariates X_i has compact support, and the conditional log-likelihood $\ell_n(\Theta)$ is continuous, twice-differentiable, and satisfies standard integrability conditions.*

Under these assumptions, we establish the consistency of the flow-based maximum likelihood estimator:

THEOREM C.2 (Consistency of the MLE). *Under the identification results of Appendix A.3 and Assumptions A.1–C.6, the maximum likelihood estimator $\hat{\Theta}_n = \arg \max_{\Theta \in \Theta} \ell_n(\Theta)$ is consistent:*

$$\|\hat{\Theta}_n - \Theta_0\|_2 = o_p(1) \quad \text{as } n \rightarrow \infty. \quad (\text{C.8})$$

PROOF. By Assumption C.5, the true latent error distribution lies within the approximating class of the conditional normalizing flow. Because the rational-quadratic spline T_ϕ^{-1} is piecewise continuous and strictly monotonic with continuous derivatives, the likelihood function $P(Y_i = j \mid X_i; \Theta)$ is continuous and bounded away from 0 and 1. Under the compactness and integrability conditions in Assumption C.6, combined with the unique nonparametric identification proved in Appendix A.3, the standard Wald conditions for the consistency of maximum likelihood estimators are satisfied, yielding $\|\hat{\Theta}_n - \Theta_0\|_2 \xrightarrow{p} 0$. $\square$

THEOREM C.3 (Asymptotic Normality and Bootstrap Validity). *Under Assumptions A.1–C.6, the MLE $\hat{\Theta}_n$ is asymptotically normal:*

$$\sqrt{n}(\hat{\Theta}_n - \Theta_0) \xrightarrow{d} \mathcal{N}(0, I(\Theta_0)^{-1}), \quad (\text{C.9})$$

where $I(\Theta_0) = -\mathbb{E}[\nabla_\Theta^2 \ell_i(\Theta_0)]$ is the positive-definite Fisher Information Matrix. Furthermore, let $g(\Theta)$ represent any smooth, continuously differentiable post-estimation functional of the distribution (such as the average marginal category-specific treatment effect $\hat{\Delta}_j$ or the Wasserstein distance $\mathcal{W}$). Then, the nonparametric bootstrap distribution of $g(\hat{\Theta}_n^*)$ is a consistent estimator of the true asymptotic distribution of $g(\hat{\Theta}_n)$.

PROOF. Since the spline function T_ϕ^{-1} is twice-differentiable everywhere, the score function $\nabla_\Theta \ell_n(\Theta)$ and Hessian $\nabla_\Theta^2 \ell_n(\Theta)$ are continuous and bounded. Thus, the Fisher information matrix is well-defined and positive-definite, establishing asymptotic normality of the MLE.

Let $g(\Theta)$ be a smooth, continuously differentiable functional mapping the estimated parameters to the causal quantities of interest. By the Delta Method, $\sqrt{n}(g(\hat{\Theta}_n) - g(\Theta_0)) \xrightarrow{d} \mathcal{N}(0, \nabla g(\Theta_0)^\top I(\Theta_0)^{-1} \nabla g(\Theta_0))$. Since standard regularity conditions for the bootstrap hold for smooth, asymptotically normal maximum-likelihood parameters in finite-dimensional spaces (Bickel and Freedman, 1981), the bootstrap distribution consistently estimates the variance of the parameters. Because the mapping $g(\cdot)$ is continuously differentiable, the bootstrap distribution of the causal estimand $g(\hat{\Theta}_n^*)$ converges in probability to the true asymptotic

distribution, validating the use of bootstrapped confidence intervals for all reported probability-based causal effects. □

D. Comprehensive Simulation Details & Full Results

D.1. Detailed Specifications of the Seven Data-Generating Processes

We evaluate finite-sample performance across seven distinct DGPs. In each setting, treatment $D_i \sim \text{Bernoulli}(0.5)$ is randomized, and baseline covariates $X_i = (X_{i1}, X_{i2})^\top$ are drawn from independent standard normal distributions. The latent thresholds partition the outcome space into $J + 1 = 5$ ordered categories: $\alpha = (-1.5, -0.5, 0.5, 1.5)^\top$.

- a. **Normal Linear (Benchmark):** The classic ordered probit setting.

$$Y_i^* = 0.5X_{i1} - 0.25X_{i2} + 0.5D_i + \varepsilon_i,$$

$$\varepsilon_i \sim \mathcal{N}(0, 1).$$

- b. **Logistic Linear (Benchmark):** The classic ordered logit setting.

$$Y_i^* = 0.5X_{i1} - 0.25X_{i2} + 0.5D_i + \varepsilon_i,$$

$$\varepsilon_i \sim \text{Logistic}(0, 1).$$

- c. **Skewed Lognormal:** Introduces heavily right-skewed latent error.

$$Y_i^* = 0.5X_{i1} - 0.25X_{i2} + 0.5D_i + \varepsilon_i,$$

$$\varepsilon_i = \exp(u_i) - \mathbb{E}[\exp(u_i)], \quad u_i \sim \mathcal{N}(0, 0.5^2).$$

- d. **Polarized Mixture (Bimodal):** Models polarized latent attitudes via a bimodal mixture of normals.

$$Y_i^* = 0.5X_{i1} - 0.25X_{i2} + 0.5D_i + \varepsilon_i,$$

$$\varepsilon_i \sim 0.5\mathcal{N}(-1.2, 0.6^2) + 0.5\mathcal{N}(1.2, 0.6^2).$$

- e. **Heteroskedastic:** Violates the proportional odds / constant scale assumption.

$$Y_i^* = 0.5X_{i1} - 0.25X_{i2} + 0.5D_i + \sigma_i\varepsilon_i,$$

$$\sigma_i = \exp(0.3D_i + 0.2X_{i1}), \quad \varepsilon_i \sim \mathcal{N}(0, 1).$$

- f. **Nonlinear Moderates:** Evaluates nonlinear latent response functions.

$$Y_i^* = 0.5 \sin(\pi X_{i1}) - 0.25X_{i2}^2 + 0.5D_i + \varepsilon_i,$$

$$\varepsilon_i \sim \mathcal{N}(0, 1).$$

- g. **High Dimensional:** Evaluates performance with 10 covariates, of which only the first four are active, under a standard normal error distribution.

D.2. Full Simulation Results Table

Table D.1 provides the complete simulation results evaluating the Wasserstein distance from the estimated conditional distribution to the true distribution across all sample sizes ($N \in \{500, 1000, 2000\}$).

TABLE D.1. Full Simulation Results: Wasserstein Distance to Truth Across All Settings

DGP Setting	N	Estimator	True	Mean	SD	Bias	RMSE
Heteroskedastic	500	Empirical	0.6049	0.6220	0.0811	0.01710	0.0825
		Ordered Probit	0.6049	0.1908	0.1138	-0.41409	0.4293
		Ordered Logit	0.6049	0.2309	0.1227	-0.37397	0.3934
		Structured Flow	0.6049	0.5826	0.1057	-0.02233	0.1075
		Model-Free Flow	0.6049	0.6077	0.0914	0.00278	0.0910
	1000	Empirical	0.6049	0.6173	0.0629	0.01247	0.0638
		Ordered Probit	0.6049	0.1654	0.0801	-0.43945	0.4466
		Ordered Logit	0.6049	0.2033	0.0874	-0.40157	0.4109
		Structured Flow	0.6049	0.5902	0.0821	-0.01465	0.0830
		Model-Free Flow	0.6049	0.6058	0.0582	0.00091	0.0579
	2000	Empirical	0.6049	0.6096	0.0397	0.00472	0.0398
		Ordered Probit	0.6049	0.1801	0.0592	-0.42478	0.4288
		Ordered Logit	0.6049	0.2208	0.0640	-0.38403	0.3893
		Structured Flow	0.6049	0.6038	0.0355	-0.00104	0.0354
		Model-Free Flow	0.6049	0.6045	0.0395	-0.00036	0.0393
High Dimensional	500	Empirical	0.7011	0.7160	0.1055	0.01483	0.1061
		Ordered Probit	0.7011	0.7052	0.1052	0.00409	0.1047
		Ordered Logit	0.7011	0.6903	0.1052	-0.01085	0.1052
		Structured Flow	0.7011	0.6640	0.1416	-0.03718	0.1458
		Model-Free Flow	0.7011	0.7044	0.1001	0.00324	0.0996
	1000	Empirical	0.7011	0.7008	0.0749	-0.00030	0.0745
		Ordered Probit	0.7011	0.6970	0.0692	-0.00412	0.0690
		Ordered Logit	0.7011	0.6822	0.0701	-0.01892	0.0723
		Structured Flow	0.7011	0.6698	0.0774	-0.03130	0.0831
		Model-Free Flow	0.7011	0.7016	0.0724	0.00046	0.0721
	2000	Empirical	0.7011	0.7078	0.0497	0.00669	0.0499
		Ordered Probit	0.7011	0.6989	0.0409	-0.00220	0.0408
		Ordered Logit	0.7011	0.6830	0.0407	-0.01815	0.0443
		Structured Flow	0.7011	0.6980	0.0516	-0.00314	0.0514
		Model-Free Flow	0.7011	0.6983	0.0458	-0.00288	0.0456
Logistic Linear	500	Empirical	0.4930	0.4712	0.1021	-0.02182	0.1039
		Ordered Probit	0.4930	0.4918	0.0836	-0.00125	0.0832
		Ordered Logit	0.4930	0.4924	0.0832	-0.00066	0.0827
		Structured Flow	0.4930	0.4965	0.0921	0.00345	0.0918
		Model-Free Flow	0.4930	0.5028	0.0978	0.00972	0.0978
	1000	Empirical	0.4930	0.4901	0.0709	-0.00294	0.0706
		Ordered Probit	0.4930	0.5102	0.0556	0.01718	0.0579
		Ordered Logit	0.4930	0.5097	0.0554	0.01669	0.0576
		Structured Flow	0.4930	0.5111	0.0685	0.01808	0.0705
		Model-Free Flow	0.4930	0.5110	0.0618	0.01797	0.0640
	2000	Empirical	0.4930	0.4923	0.0549	-0.00075	0.0546
		Ordered Probit	0.4930	0.5044	0.0427	0.01138	0.0440
		Ordered Logit	0.4930	0.5055	0.0415	0.01251	0.0431
		Structured Flow	0.4930	0.5039	0.0476	0.01089	0.0486
		Model-Free Flow	0.4930	0.5047	0.0521	0.01170	0.0531
Nonlinear Moderates	500	Empirical	0.7348	0.7374	0.1066	0.00254	0.1061
		Ordered Probit	0.7348	0.7234	0.1067	-0.01141	0.1068
		Ordered Logit	0.7348	0.7364	0.1056	0.00159	0.1051
		Structured Flow	0.7348	0.7303	0.1255	-0.00454	0.1250
		Model-Free Flow	0.7348	0.7386	0.0973	0.00373	0.0969
	1000	Empirical	0.7348	0.7516	0.0795	0.01678	0.0809
		Ordered Probit ₁₃	0.7348	0.7337	0.0786	-0.00111	0.0782
		Ordered Logit	0.7348	0.7421	0.0783	0.00730	0.0783
		Structured Flow	0.7348	0.7534	0.0952	0.01862	0.0966
		Model-Free Flow	0.7348	0.7437	0.0700	0.00885	0.0702

E. Empirical Application Details & Full Tables

This section reports the full set of empirical results corresponding to the two applications discussed in Section 5: Tomz and Weeks (2020) and Mattingly et al. (2025). Standard errors are calculated using $B = 500$ bootstrap replications and are presented in parentheses.

E.1. Mattingly et al. (2025): Persuasive Effects of Authoritarian Propaganda

TABLE E.1. Empirical Application Estimates: Support for Political Model (Mattingly et al., 2025)

Metric / Category	Empirical	Ordered Probit	Ordered Logit	Structured Flow	Model-Free Flow
Wasserstein Unit	1.073 (0.038)	0.974 (0.048)	0.986 (0.047)	0.917 (0.116)	1.011 (0.246)
<i>Category Effects (AME)</i>					
Category 1	-0.138 (0.011)	-0.171 (0.007)	-0.166 (0.006)	-0.113 (0.024)	-0.092 (0.050)
Category 2	-0.135 (0.010)	-0.095 (0.006)	-0.113 (0.006)	-0.120 (0.012)	-0.137 (0.028)
Category 3	-0.085 (0.013)	0.015 (0.001)	0.013 (0.002)	-0.055 (0.016)	-0.078 (0.015)
Category 4	0.152 (0.011)	0.071 (0.003)	0.086 (0.004)	0.098 (0.014)	0.063 (0.025)
Category 5	0.110 (0.009)	0.075 (0.004)	0.086 (0.004)	0.096 (0.014)	0.105 (0.022)
Category 6	0.097 (0.008)	0.105 (0.006)	0.094 (0.005)	0.094 (0.031)	0.139 (0.029)
<i>Cumulative Effects</i>					
$P(Y \geq 2)$	0.138 (0.011)	0.171 (0.007)	0.166 (0.006)	0.113 (0.024)	0.092 (0.050)
$P(Y \geq 3)$	0.273 (0.013)	0.266 (0.012)	0.279 (0.012)	0.233 (0.026)	0.229 (0.074)
$P(Y \geq 4)$	0.359 (0.012)	0.251 (0.013)	0.267 (0.014)	0.288 (0.023)	0.307 (0.070)
$P(Y \geq 5)$	0.206 (0.010)	0.180 (0.010)	0.180 (0.010)	0.190 (0.024)	0.244 (0.049)
$P(Y \geq 6)$	0.097 (0.008)	0.105 (0.006)	0.094 (0.005)	0.094 (0.031)	0.139 (0.029)

Table E.1 reports the point estimates and bootstrapped standard errors for average marginal category-specific effects and cumulative effects across the five candidate estimators.

E.2. Tomz and Weeks (2020): Human Rights and Public Support for Intervention

TABLE E.2. Empirical Application Estimates: Support for Strike (Tomz and Weeks, 2020)

Metric / Category	Empirical	Ordered Probit	Ordered Logit	Structured Flow	Model-Free Flow
Wasserstein Unit	0.513 (0.065)	0.500 (0.058)	0.507 (0.059)	0.504 (0.107)	0.509 (0.077)
<i>Category Effects (AME)</i>					
Category 1	0.084 (0.017)	0.081 (0.009)	0.078 (0.009)	0.073 (0.018)	0.068 (0.018)
Category 2	0.068 (0.022)	0.065 (0.008)	0.071 (0.009)	0.080 (0.019)	0.092 (0.021)
Category 3	0.017 (0.020)	0.015 (0.002)	0.016 (0.002)	0.019 (0.017)	0.023 (0.019)
Category 4	-0.061 (0.025)	-0.048 (0.006)	-0.053 (0.007)	-0.065 (0.022)	-0.084 (0.025)
Category 5	-0.108 (0.022)	-0.113 (0.013)	-0.113 (0.013)	-0.106 (0.031)	-0.098 (0.024)
<i>Cumulative Effects</i>					
$P(Y \geq 2)$	-0.084 (0.017)	-0.081 (0.009)	-0.078 (0.009)	-0.073 (0.018)	-0.068 (0.018)
$P(Y \geq 3)$	-0.152 (0.024)	-0.146 (0.017)	-0.150 (0.017)	-0.153 (0.032)	-0.160 (0.028)
$P(Y \geq 4)$	-0.169 (0.025)	-0.161 (0.019)	-0.166 (0.019)	-0.172 (0.038)	-0.183 (0.027)
$P(Y \geq 5)$	-0.108 (0.022)	-0.113 (0.013)	-0.113 (0.013)	-0.106 (0.031)	-0.098 (0.024)

Table E.2 reports the results for the replication of Tomz and Weeks (2020), indicating consistency across the parametric and flexible flow-based estimators.

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